



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PART TEST

04-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,B,C	A,B,C,D	A,B,D	A,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	125.00	0.50	6.28 to 6.30	45.00	5.00	84.50 to 85.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	4	3	5	2	2	9

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,C,D	A,B,C	B,C,D	A,C	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.14	66.66 to 66.67	8.87 to 8.88	10.66 to 10.67	1.33 to 1.34	9.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	1	9	6	4	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,C	A,B	A,C	A,B,C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	10.00	0.50	4.00	1.33	1.50	0.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	2	3	0	1	6

HINT – SHEET
PART-1 : PHYSICS
SECTION-I
1. Ans (B,C)

$$y = 2e^{-4(x^2 + 81t^2 + 18xt + 3/4)}$$

$$y = 2e^{-4\{(x+9t)^2 + 3/4\}}$$

wave speed = 9 m/s (negative x-direction)

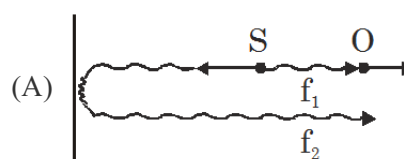
$$v = \sqrt{\frac{I}{\mu}} \Rightarrow T = \mu v^2 = 0.3 \times (9)^2 = 0.3 \times 81$$

$$= 24.3 \text{ N}$$

3. Ans (A,B,C,D)

$$\beta = 10 \log \frac{I}{I_0}, I \propto \frac{1}{r^2}$$

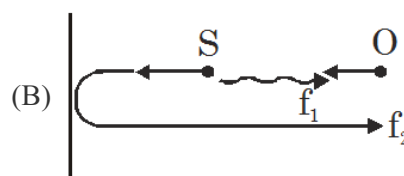
 If r increases from r to 10r, intensity decreases 100 times $\Rightarrow$ b decreases by 20dB and intensity decreases by a factor of I_w .

4. Ans (A,B,D)


$$f_1 = 360 \left(\frac{330 - 10}{330 + 30} \right) = 320 \text{ Hz}$$

$$f_2 = 360 \left(\frac{330 - 10}{330 - 30} \right) = 384 \text{ Hz}$$

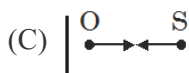
$$\therefore \text{beat frequency} = 384 - 320 = 64 \text{ Hz}$$



$$f_1 = 360 \left(\frac{330 + 10}{330 + 30} \right) = 340 \text{ Hz}$$

$$f_2 = 360 \left(\frac{330 + 10}{330 - 30} \right) = 408 \text{ Hz}$$

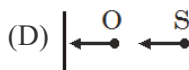
$$\therefore \text{beat frequency} = 408 - 340 = 68 \text{ Hz}$$



$$f_1 = 360 \left(\frac{330 + 10}{330 - 30} \right) = 408 \text{ Hz}$$

$$f_2 = 360 \left(\frac{330 - 10}{330 - 30} \right) = 384 \text{ Hz}$$

$$\text{beat frequency} = 408 - 384 = 24 \text{ Hz}$$



$$f_1 = 360 \left(\frac{330 - 10}{330 - 30} \right) = 384 \text{ Hz}$$

$$f_2 = 360 \left(\frac{330 + 10}{330 - 30} \right) = 408 \text{ Hz}$$

$$\text{beat frequency} = 408 - 384 = 24 \text{ Hz}$$

5. Ans (A,C,D)

For central maxima

$$\Delta x = 0$$

$$\mu_w \frac{y d}{D} + (\mu - \mu_w) t_2 - (\mu - \mu_w) t_1 = 0$$

$$y \frac{d}{D} = \frac{(\mu - \mu_w)(t_1 - t_2)}{\mu_w}$$

$$y = \frac{\left(\frac{1.4}{4/3} - 1 \right) (1.25) \times 10^{-6} \times 80 \times 10^{-2}}{2 \times 10^{-3}}$$

$$= 0.025 \text{ mm}$$

(C) Optical path difference at point C

$$\Delta x = (\mu - \mu_w)(t_1 - t_2)$$

$$= \left(1.4 - \frac{4}{3} \right) (2.5 - 1.25) \times 10^{-6} = \frac{1}{12} \times 10^{-6}$$

$$\Delta \phi = \frac{2\pi}{\pi} \cdot \Delta x$$

$$= \frac{2\pi}{5000 \times 10^{-10}} \times \frac{1}{12} \times 10^{-6}$$

$$= \left(\frac{\pi}{3} \right)$$

$$I_r = 4I_0 \cdot \cos^2 \left(\frac{\phi}{2} \right)$$

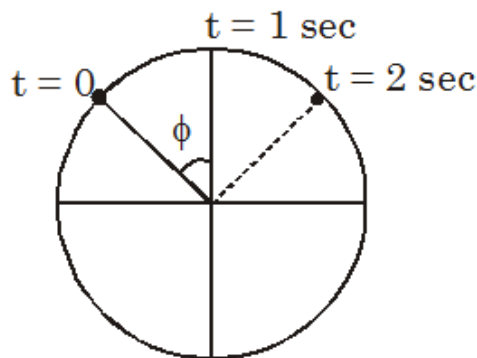
$$\frac{I_r}{4I_0} = \frac{3}{4}$$

$$I_r = 75\%$$

$$(D) \text{ fringe width} = \frac{\lambda D}{\mu_w d}$$

$$= 0.15 \text{ mm}$$

6. Ans (A,B,C)



$$T = 6 \text{ sec}$$

$$\omega = \frac{2\pi}{T} = \frac{\pi}{3} \text{ rad/sec}$$

$$\text{If } x = A \sin \omega t$$

$$v = A \omega \cos \omega t$$

$$\frac{1}{4} = A \times \frac{\pi}{3} \cos \frac{\pi}{3} \times 1 \quad \Delta t = 1 \text{ sec}$$

$$\frac{1}{4} = A \cdot \frac{\pi}{3} \cdot \frac{1}{2}$$

$$A = \frac{3}{2\pi} \text{ m}$$

$$\phi = \omega t = \frac{\pi}{3} \times 1 = \frac{\pi}{3}$$

$$v = A \omega \cos \omega t$$

$$= \frac{3}{2\pi} \times \frac{\pi}{3} \cos \left(\frac{\pi}{3} \times 5 \right) = \frac{1}{2} \cos \left(\pi - \frac{2\pi}{3} \right) = 0.25 \text{ m/s}$$

$$\text{acc. } a = -A \omega^2 \sin \omega t$$

$$= -\frac{3}{2\pi} \times \frac{\pi^2}{9} \sin \frac{\pi}{3} \times 5$$

$$= +\frac{\pi}{6} \times \frac{\sqrt{3}}{2} = \frac{\pi}{4\sqrt{3}} \text{ m/s}^2$$

PART-1 : PHYSICS

SECTION-II

1. Ans (125.00)

$$x = 3 \sin(100t) + \frac{8(1 + \cos 100t)}{2}$$

$$(x - 4) = 3 \sin 100t + 4 \cos(100t)$$

$$\text{Amplitude} = 5 \text{ m, } \omega = 100 \text{ rad/s}$$

$$T E = \frac{1}{2} m \omega^2 A^2$$

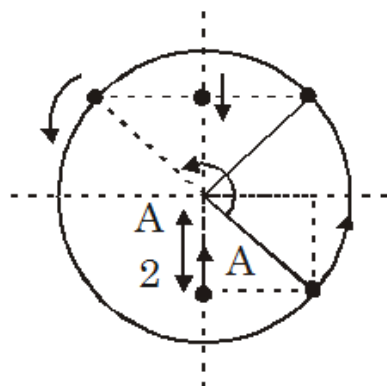
$$= \frac{1}{2} \times 1 \times 10^4 \times 25$$

$$= 5 \times 10^3 \times 25$$

$$= 125 \times 10^3 \text{ J}$$

$$= 125 \text{ kJ}$$

2. Ans (0.50)



Using concept of SHM

0.4 = half of the period

$$0.4 = \frac{T}{2} \Rightarrow T = 0.8$$

$$\ell = 0.6$$

$$f = \left(\frac{5}{4} \right)$$

$$\text{Also } \frac{3\lambda}{2} = 0.6,$$

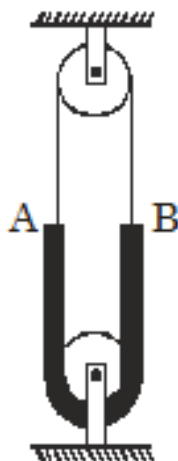
$$\lambda = 0.4 \text{ m, } v = f\lambda = \frac{5}{4} \times 0.4 = 0.5 \text{ m/s}$$

$$= 0.50 \text{ m/s}$$

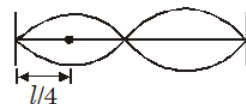
3. Ans (6.28 to 6.30)

$$T = 2\pi\sqrt{\frac{\ell}{2g}} = 2\pi\sqrt{\frac{20}{2 \times 10}} = 2\pi$$

$$T = 2 \times 3.14 = 6.28 \text{ sec}$$



4. Ans (45.00)



$$\frac{\ell}{4} = \frac{\lambda}{4}$$

$$\ell = \lambda$$

$$v = f\lambda$$

$$\sqrt{\frac{T_1}{\mu}} = f_1 \ell \dots\dots\dots(1)$$



$$\frac{3\lambda}{4} = \frac{\ell}{4}$$

$$\lambda = \frac{\ell}{3}$$

$$v = f\lambda$$

$$\sqrt{\frac{T_2}{\mu}} = f_2 \cdot \frac{\ell}{3} \dots\dots\dots(2)$$

$$\sqrt{\frac{T_1}{T_2}} = \frac{3f_1}{f_2}$$

$$f_2 = 3 \times 30 \times \sqrt{\frac{25}{100}}$$

$$= 45 \text{ Hz}$$

5. Ans (5.00)

$$(i) 2t_1 (\mu_1 - 1) = 4\lambda$$

$$t_1 = 2\mu\text{m}$$

$$(ii) 2t_1 \left(\frac{\mu_1}{\mu_1} - 1 \right) + t_2 \left(\frac{1}{\mu_1} - 1 \right) = 0$$

$$t_2 = 1\mu\text{m}$$

$$a = 2t_1 + t_2 = 5\mu\text{m}.$$

6. Ans (84.50 to 85.00)

$$\text{Intensity} = \frac{\Delta P^2}{2\rho v}$$

$$= \frac{0.5 \times 0.5}{2 \times 1.25 \times 330}$$

$$= \frac{1}{3300} \text{ W/m}^2$$

$$L = \log_{10} \left(\frac{I}{I_0} \right)$$

$$= \log_{10} \left(\frac{1/3300}{10^{-12}} \right)$$

$$= \log_{10} \left(\frac{10^{10}}{33} \right)$$

$$= 10 - \log_{10} 33$$

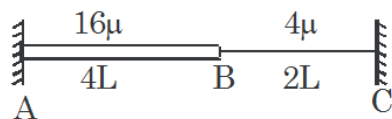
$$= 10 - 1.52$$

$$= 8.48 \text{ bel} = 84.8 \text{ decibel}$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (4)**



$$\frac{nv_1}{2 \times 4L} = \frac{mv_2}{2 \times 2L}$$

$$\frac{n}{m} = \frac{2v_2}{v_1} = \frac{2\sqrt{\frac{T}{4\mu}}}{\sqrt{\frac{T}{16\mu}}}$$

$$= 2 \times 2 = \frac{4}{1}$$

$$\frac{n}{m} = \frac{4}{1}$$

Minimum $n = 4$

2. **Ans (3)**

$$0.6 + Le = \frac{\lambda}{4}$$

$$2.2 + Le = \frac{3\lambda}{4}$$

$$\lambda = 3.2 \text{ m}$$

$$0.6 + Le = 0.8$$

$$Le = 0.2 \text{ m}$$

$$Le = 0.2 \text{ m} = 0.6 r$$

$$r = \left(\frac{1}{3}\text{m}\right)$$

3. **Ans (5)**

$$\text{Path different} = d\sin\theta \text{ and } d\tan\theta = \frac{dh}{D}$$

$$\text{time gap} = \frac{dh}{DV} = \frac{2 \times 10^{-3} \times 10 \times 10^{-3}}{300 \times 1\text{m}}$$

$$= \frac{1}{150} \mu\text{s}$$

$$m = 150, \frac{150}{30} = 5$$

4. **Ans (2)**

From conservation of angular momentum

$$mv\frac{R}{2} + mv\frac{R}{2} + 0 = \frac{5}{2}mR^2W$$

$$W = \left(\frac{2v}{5R}\right)$$

$$\text{Angular freq. } W_0 = \sqrt{\frac{C}{I}} = \sqrt{\frac{2C}{5mR^2}}$$

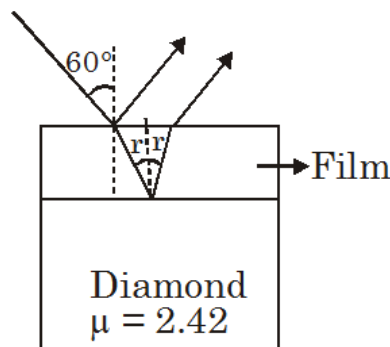
$$\therefore \text{Angular amplitude}(\beta) = \left(\frac{W}{W_0}\right)$$

$$= \frac{2v}{5R} \sqrt{\frac{5mR^2}{2C}}$$

$$= V \sqrt{\frac{2}{5} \frac{m}{C}}$$

$$= 20 \sqrt{\frac{2}{5} \times \frac{2}{80}} = 2 \text{ radian}$$

5. **Ans (2)**



From Snell's law

$$1. \sin 60^\circ = \sqrt{3} \sin r$$

$$r = 30^\circ$$

Optical path difference (Δx)

$$\begin{aligned} \Delta x &= 2\mu t \sec r - 2t \tan r \sin i \\ &= 2\sqrt{3} (t \sec 30^\circ) - 2t \tan 30^\circ \cdot \sin 60^\circ \\ &= 3t \end{aligned}$$

For constructive interference

$$3t = \lambda$$

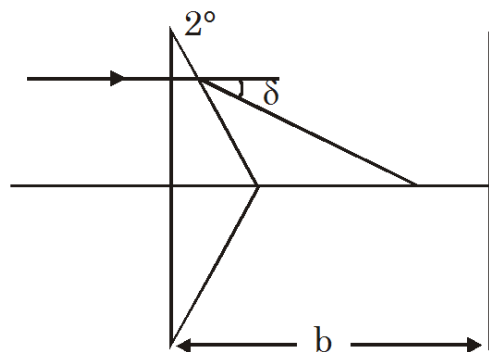
$$t = \frac{\lambda}{3} = \frac{600}{3} = 200\text{nm}$$

$$= 2 \times 10^{-7}\text{m}$$

$$= 0.2 \mu\text{m}$$

$$10t = 2 \mu\text{m}$$

6. Ans (9)



$$\text{Fringe width} = \frac{\lambda}{2(\mu_g - \mu_w)A}$$

$$= \frac{500 \times 10^{-9} \times 180}{2 \times \left(\frac{3}{2} - \frac{4}{3}\right) \times 2 \times \pi}$$

$$= \frac{5 \times 18 \times 10^{-6} \times 6}{4\pi} = 45\mu\text{m}$$

$$\therefore \frac{w}{5} = 9$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A,B,D)

The solvated e^- occupy cavities which leads to increase in volume. On standing it forms H_2 gas.

2. Ans (A,C,D)

Solubility of oxalate, hydroxide and fluorides increases from Ca to Ba but solubility of carbonate decreases.

3. Ans (A,B,C)

2-Aminobutane-1, 4-dioic anhydride is the correct IUPAC name.

6. Ans (A,B,D)

$$r = K(B_2)(A)$$

using first equation

$$K_{eq} = \frac{(A)^2}{(A_2)}$$

$$\therefore (A) = K_{eq}^{1/2} \cdot (A_2)^{1/2}$$

$$\Rightarrow r = K(K_{eq})^{1/2} (B_2)(A_2)^{1/2}$$

$$= K'(A_2)^{1/2}(B_2)^1$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (3.14)

$$\lambda = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda}$$

$$\frac{dp}{d\lambda} = -\frac{h}{\lambda^2}$$

$$\therefore \Delta p = \frac{h}{\lambda^2} \cdot \Delta \lambda$$

$$\Rightarrow \Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Rightarrow \Delta x \cdot \frac{h}{\lambda^2} \cdot \Delta \lambda \geq \frac{h}{4\pi}$$

$$\Rightarrow \Delta x \cdot \Delta \lambda \geq \frac{\lambda^2}{4\pi} \dots \dots (1)$$

$$n = 3, z = 2$$

$$2\pi \cdot r_0 \times \frac{9}{2} = 3 \cdot \lambda$$

$$\lambda = 1.5 \pi \text{Å}$$

$$\Delta x \cdot \Delta \lambda \geq \frac{9}{4} \frac{\pi^2}{4\pi} \Rightarrow \Delta \lambda \geq \pi$$

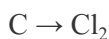
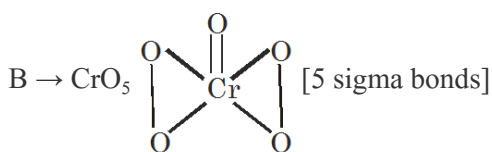
2. Ans (66.66 to 66.67)

$$\frac{Y}{Z} = \frac{3\lambda_1}{4\lambda_2} = \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$$

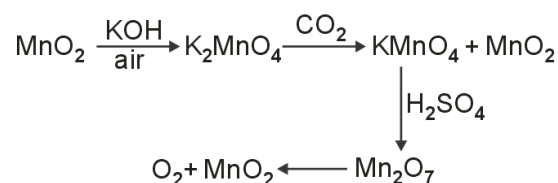
$$\%Z = \frac{2}{3} \times 100 = 66.67\%$$

3. Ans (8.87 to 8.88)

A $\rightarrow CrO_2 \cdot Cl_2$ [hybridisation is d_3s . The orbitals involved are d_{xy} , d_{yz} and d_{zx}]



4. Ans (10.66 to 10.67)



$$\therefore x = 3, y = 32$$

5. Ans (1.33 to 1.34)

$$\frac{7-3}{3} = \frac{4}{3} = 1.333$$

(i), (ii), (iii), (iv), (vi), (ix), (x) less than 6.4.

6. Ans (9.00)

$$x = 12, y = 3$$

PART-2 : CHEMISTRY

SECTION-III

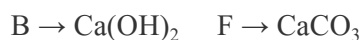
1. Ans (5)

(d), (e), (f), (g), (h)

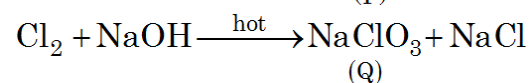
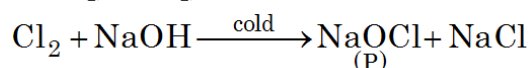
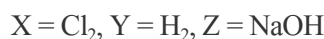
2. Ans (1)

(e)

3. Ans (9)



4. Ans (6)



Sum of oxidation state of 'Cl' in P & Q is $1 + 5 = 6$

5. Ans (4)

$$(1)^2 \left[\frac{1}{1^2} - \frac{1}{4^2} \right] = (1)^2 \left(\frac{1}{(n_1/Z)^2} - \frac{1}{(n_2/Z)^2} \right)$$

For Be^{+3} , $Z = 4$, $16 \rightarrow 4$

For H its $1 \rightarrow 4$

$$2\pi \left[0.5 \times \frac{4^2}{1} \right] = 4 \cdot \lambda$$

$$\lambda = 4\pi \text{ \AA}$$

6. Ans (3)

$$\frac{d}{dt}(\ln k) = \frac{E_a}{RT^2}$$

$$0 + \frac{5000}{T^2} + \frac{3}{3T} \cdot 3 = \frac{E_a}{RT^2}$$

$$5000 + 3T = \frac{E_a}{R}$$

$$E_a = 11.8 \text{ kcal/mol}$$

$$T = 300 \text{ K}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B,C)

$$e^x f(x) = 2(f(x))^2 - 3x$$

$$e^x f(x) + e^x f'(x) = 4f(x)f'(x) - 3$$

$$f'(x)(e^x - 4f(x)) = -3 - e^x \cdot f(x)$$

$$f'(x) = \frac{-3 - e^x f(x)}{e^x - 4f(x)}$$

$$\int \frac{e^x(2e^{2x} - 8e^x f(x) + f(x)) + 3}{(e^x - 4f(x))(e^{2x} - f(x))^2} dx$$

$$\int \frac{2e^{3x} - 8e^{2x} f(x) + 3 + e^x f(x)}{(e^x - 4f(x))(e^{2x} - f(x))^2} dx$$

$$\int \frac{2e^{2x} - \frac{-3 - e^x f(x)}{e^x - 4f(x)}}{(e^{2x} - f(x))^2} dx$$

$$\int \frac{2e^{2x} - f'(x)}{(e^{2x} - f(x))^2} dx = \frac{-1}{e^{2x} - f(x)} + c$$

Put $x = 1$

$$e f(1) = 2(f(1))^2 - 3$$

$$\Rightarrow f(1) = \frac{e + \sqrt{e^2 + 24}}{4}$$

$$\int_0^1 g(x) dx = \left[\frac{-1}{e^{2x} - f(x)} \right]_0^1$$

$$= 2 \left(\frac{2}{e + \sqrt{e^2 + 24} - 4e^2} + 1 \right)$$

2. Ans (A,C)

$$I_1 = \int_0^{\ln 10^4} e^x \{e^x\} dx$$

$$e^x = t \Rightarrow e^x dx = dt$$

$$= \int_1^{10^4} \{t\} dt = (10^4 - 1) \int_0^1 t dt = \frac{9999}{2}$$

$$I_2 = \int_0^{9 \times 10^6} \frac{\{\sqrt{x}\}}{\sqrt{x}} dx$$

$$\sqrt{x} = t \Rightarrow \frac{dx}{\sqrt{x}} = 2dt$$

$$= \int_0^{3000} 2\{t\} dt$$

$$= 6000 \int_0^1 t dt = 3000$$

3. Ans (A,B)

$$S_k = \int_0^1 x^2(1-x)^k dx = \int_0^1 (1-x)^2 x^k dx$$

$$= \int_0^1 (x^k - 2x^{k+1} + x^{k+2}) dx$$

$$= \frac{1}{k+1} - \frac{2}{k+2} + \frac{1}{k+3}$$

$$S_k = \left(\frac{1}{k+1} - \frac{1}{k+2} \right) - \left(\frac{1}{k+2} - \frac{1}{k+3} \right)$$

$$\therefore \lim_{n \rightarrow \infty} \sum_{k=1}^{\infty} S_k = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

4. Ans (A,C)

$$\int \frac{(\sin x) e^{\sqrt{\cos x}}}{\sqrt{\cos x}} dx \quad \text{Let } \cos x = t^2$$

$$-\sin x dx = 2t dt$$

$$\sin x dx = -2t dt$$

$$-2 \int \frac{e^t \cdot t dt}{t} = -2e^{\sqrt{\cos x}} + c$$

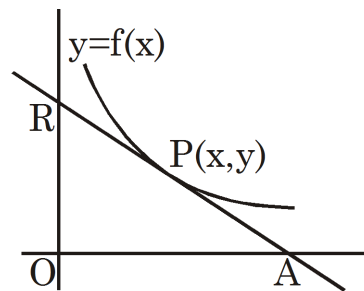
$$f(0) = -2e$$

5. Ans (A,B,C,D)

$$\text{Equation of tangent } Y - y = \frac{dy}{dx}(X - x)$$

$$\therefore A = \left(x - y \frac{dx}{dy}, 0 \right)$$

$$B = \left(0, y - x \frac{dy}{dx} \right)$$



$$\text{Thus } \frac{1}{2} \left(x - y \frac{dx}{dy} \right) \left(y - x \frac{dy}{dx} \right) = 2$$

$$x^2 \left(\frac{dy}{dx} \right)^2 - 2xy \frac{dy}{dx} + y^2 = -4 \frac{dy}{dx} \quad \dots(1)$$

Diff. w.r.t. x

$$2x \left(\frac{dy}{dx} \right)^2 + 2x^2 \left(\frac{dy}{dx} \right) \frac{d^2y}{dx^2} - 2y \frac{dy}{dx} - 2x \left(\frac{dy}{dx} \right)^2$$

$$- 2xy \frac{d^2y}{dx^2} + 2y \frac{dy}{dx} = -4 \frac{d^2y}{dx^2}$$

$$\therefore \frac{d^2y}{dx^2} = 0 \text{ or } x^2 \frac{dy}{dx} - xy + 2 = 0$$

$$\frac{d^2y}{dx^2} = 0 \text{ or } \frac{dy}{dx} - \frac{1}{x} \cdot y = \frac{-2}{x^2}$$

$$\frac{dy}{dx} = \lambda \text{ I.F.} = e^{\int -\frac{1}{x} dx} = \frac{1}{x}$$

$$\text{Put } \frac{dy}{dx} = \lambda \text{ in (1)}$$

$$\therefore y \cdot \frac{1}{x} = \int \frac{2}{x^3} dx$$

$$x^2 \lambda^2 - 2xy \lambda + y^2 = -4\lambda \quad \frac{y}{x} = \frac{+1}{x^2} + c$$

$$(1, 1) \quad (1, 1)$$

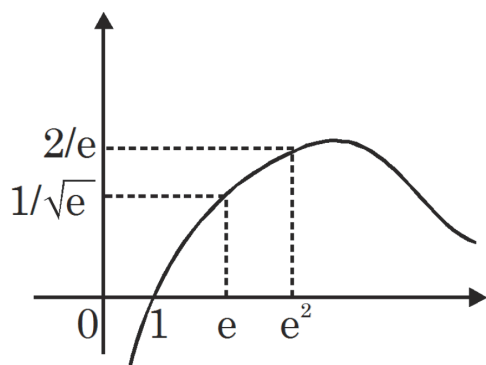
$$\therefore x^2 + 2xy + y^2 = 4 \quad \therefore c = 0$$

$$x + y = 2$$

$$xy = 1$$

6. Ans (A,B,C,D)

$$f'(x) = \frac{2 - \ln x}{2x^{3/2}} = 0, \quad x = e^2$$



$$A = \int_1^{e^2} \frac{\ln x}{\sqrt{x}} dx < (e^2 - 1) \frac{2}{e}$$

$$\therefore A < 2 \left(e - \frac{1}{e} \right)$$

$$\text{Now } A = \int_1^{e^2} \frac{\ln x}{\sqrt{x}} dx$$

$$= \int_1^e \frac{\ln x}{\sqrt{x}} dx + \int_e^{e^2} \frac{\ln x}{\sqrt{x}} dx$$

$$A < (e - 1) \frac{1}{\sqrt{e}} + (e^2 - e) \frac{2}{e}$$

$$A > \frac{1}{2} (e^2 - 1) \times \frac{2}{e}$$

$$A < (e - 1) \left(2 + \frac{1}{\sqrt{e}} \right) \quad A > e - \frac{1}{e}$$

$$\text{Also } A = \int_1^{e^2} \frac{\ln x}{\sqrt{x}} dx$$

$$\therefore 3 < A < 5$$

But $\ln x = t$

$$A = \int_0^2 t \sqrt{e^t} dt$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (10.00)

Differentiating with respect to x .

$$\begin{aligned} \frac{x^4 + 1}{x(x^2 + 1)^2} &= \frac{a}{x} - \frac{2bx}{(1 + x^2)^2} \\ &= \frac{ax^4 + (2a - 2b)x^2 + a}{x \cdot (1 + x^2)^2} \end{aligned}$$

comparing numerator, we get $a = b = 1$

$$\therefore \text{Given limit} = \lim_{x \rightarrow 0} \frac{\sin 10x + \cos x - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 10x}{x} + \left(\frac{\cos x - 1}{x^2} \right) x = 10 + 0 = 10$$

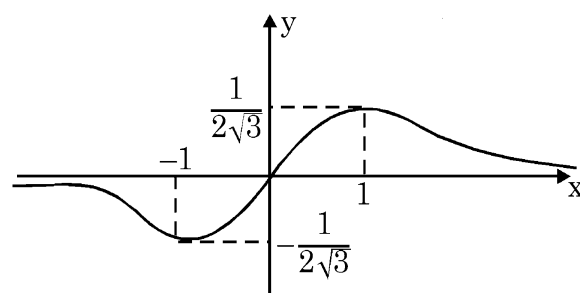
2. Ans (0.50)

Applying Leibniz

$$x^2 f(x) = \frac{x}{\sqrt{3}} - f(x)$$

$$\Rightarrow f(x) = \frac{x}{\sqrt{3}(x^2 + 1)}$$

$$f'(x) = \frac{1}{\sqrt{3}} \left(\frac{(x^2 + 1) - (2x)x}{(x^2 + 1)^2} \right) = \frac{1 - x^2}{\sqrt{3}(x^2 + 1)^2}$$

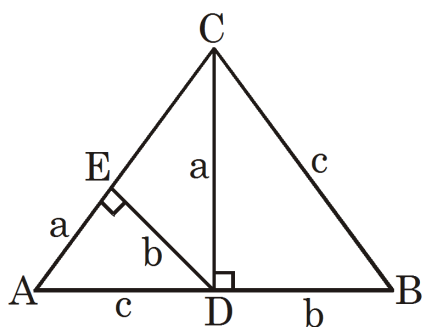


$$\therefore k \in \left(-\frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}} \right) - \{0\}$$

$$a\sqrt{3} = 0.5 \quad b = 0$$

$$a\sqrt{3} + b = 0.5$$

3. Ans (4.00)



$$\text{Area of } \triangle ABC = \frac{1}{2}a(b+c)$$

$$\text{Area of } \triangle ADE = \frac{1}{2}ab$$

$$\therefore \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle ADE} = \frac{a(b+c)}{ab} = 1 + \frac{c}{b}$$

$$\text{We know that } a^2 + b^2 = c^2$$

We also see that $\triangle ADE \sim \triangle DCB$

$$\therefore \frac{c}{a} = \frac{a}{b} \Rightarrow a^2 = bc$$

$$\therefore bc + b^2 = c^2$$

$$\Rightarrow c^2 - bc - b^2 = 0$$

$$\Rightarrow \left(\frac{c}{b}\right)^2 - \left(\frac{c}{b}\right) - 1 = 0$$

$$\Rightarrow \frac{c}{b} = \frac{1 \pm \sqrt{5}}{2}$$

$$\Rightarrow \frac{c}{b} = \frac{\sqrt{5} + 1}{2}$$

$$\therefore \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle ADE} = 1 + \frac{\sqrt{5} + 1}{2}$$

$$= \frac{3 + \sqrt{5}}{2}$$

$$\therefore a = 3, b = 5, c = 2$$

4. Ans (1.33)

$$f'(x) = 3 + \int_0^x \frac{1 + \sin t}{t^2 + 2} dt \quad \forall x \in \mathbb{R} \Rightarrow f(0) = 3$$

$$f'(x) = \frac{1 + \sin x}{x^2 + 2} \Rightarrow f'(0) = \frac{1}{2}$$

$$f''(x) = \frac{(x^2 + 2) \cos x - 1(1 + \sin x) \times 2x}{(x^2 + 2)^2}$$

Let $P(x) = ax^2 + bx + 3$. Hence

$$P'(x) = 2ax + b \Rightarrow b = \frac{1}{2} \text{ and } P''(x) = 2a = \frac{1}{2} \Rightarrow a = \frac{1}{4}$$

$$\therefore P(x) = \frac{1}{4}x^2 + \frac{x}{2} + 3 = \frac{1}{4}(x^2 + 2x + 12)$$

$$= \frac{1}{4}[(x+1)^2 + 11]$$

$$P'(x) = \frac{x+1}{2} \Rightarrow P'(-2) = -\frac{1}{2}$$

Equation of tangent at $(-2, 3)$ is

$$y - 3 = -\frac{1}{2}(x + 2)$$

$$\text{Put } x = 0, y = 3 - 1 = 2$$

$$y = -\frac{x}{2} + 2$$

$$\text{Area} = \int_{-2}^0 \left(\frac{x^2 + 2x + 12}{4} - \left(2 - \frac{x}{2}\right) \right) dx$$

$$= \int_{-2}^0 \frac{x^2 + 4x + 4}{4} dx = \frac{1}{4} \times \int_{-2}^0 (x+2)^2 dx$$

$$= \frac{1}{12} \times [(x+2)^3]_{-2}^0 = \frac{8}{12} = \frac{2}{3}$$

$$A = \frac{2}{3} \Rightarrow 2A = \frac{4}{3} = 1.33$$

5. Ans (1.50)

$$I_n = \int_0^1 e^{2x} (x-1)^n dx$$

$$I_n = \left[(x-1)^n \cdot \frac{e^{2x}}{2} \right]_0^1 - \int_0^1 n(x-1)^{n-1} \frac{e^{2x}}{2} dx$$

$$I_n = \frac{-(-1)^n}{2} - \frac{n}{2} I_{n-1}$$

$$I_0 = \int_0^1 e^{2x} dx = \left[\frac{e^{2x}}{2} \right]_0^1 = \frac{e^2 - 1}{2}$$

$$I_1 = \frac{1}{2} - \frac{1}{2} \frac{(e^2 - 1)}{2} = \frac{3 - e^2}{4}$$

$$I_2 = -\frac{1}{2} - \left(\frac{3 - e^2}{4} \right) = \frac{e^2 - 5}{4}$$

$$\therefore n = 2$$

$$I_1 = \int_{\sec^2 z}^{2-\tan^2 z} x f(x(3-x)) dx$$

$$I_1 = \int_{\sec^2 z}^{2-\tan^2 z} (3-x) f((3-x)x) dx$$

$$I_1 = 3 \int_{\sec^2 z}^{2-\tan^2 z} f((3-x)x) dx - I_1$$

$$I_1 = 3I_2 - I_1$$

$$\Rightarrow 2I_1 = 3I_2$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{3}{2}$$

6. Ans (0.50)

$$2y e^{y^2} \frac{dy}{dx} + 2x e^{y^2} = 2x \sin x^2$$

$$\text{Put } e^{y^2} = t \Rightarrow 2y e^{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + 2x \cdot t = 2x \sin x^2$$

$$\text{Integrating factor} = e^{\int 2x dx} = e^{x^2}$$

$$t \cdot e^{x^2} = \int 2x \sin x^2 e^{x^2} dx$$

$$\text{Put } x^2 = z \Rightarrow 2x dx = dz$$

$$t \cdot e^{x^2} = \int e^z \sin z dz$$

$$t \cdot e^{x^2} = \frac{e^z (\sin z - \cos z)}{2} + c$$

$$\Rightarrow e^{x^2+y^2} = \frac{e^{x^2} (\sin x^2 - \cos x^2)}{2} + c$$

$$e^0 = \frac{e^0 (0-1)}{2} + c$$

$$\Rightarrow 1 = -\frac{1}{2} + c \Rightarrow c = \frac{3}{2}$$

$$\therefore f(x) = e^{-x^2}$$

$$f(\sqrt{\ln 2}) = e^{-\ln 2} = e^{\ln \frac{1}{2}} = \frac{1}{2}$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (6)

$$f(x) + f(x+1) + \dots + f(x+10)$$

$$= |e^x - 1| + |e^x - 2| + \dots + |e^x - 11|$$

$$\int_0^{11} f(x) dx = \int_0^1 f(x) dx + \int_1^2 f(x) dx + \dots + \int_{10}^{11} f(x) dx$$

$$\text{Put } x = t + 1 \quad x = t + 10$$

$$\int_0^{11} f(x) dx = \int_0^1 f(x) dx + \int_0^1 f(x+1) dx + \dots + \int_0^1 f(x+10) dx$$

$$= \int_0^1 [f(x) + f(x+1) + \dots + f(x+10)] dx$$

$$= \int_0^1 (|e^x - 1| + |e^x - 2| + \dots + |e^x - 11|) dx$$

$$= \int_0^1 (e^x - 1) dx + \int_0^{\ln 2} (2 - e^x) dx + \int_{\ln 2}^1 (e^x - 2) dx$$

$$+ \int_0^1 (3 - e^x) dx + \dots + \int_0^1 (11 - e^x) dx$$

$$= [e^x - x]_0^1 + [2x - e^x]_0^{\ln 2} + [e^x - 2x]_{\ln 2}^1$$

$$+ [3x - e^x]_0^1 + \dots + [11x - e^x]_0^1$$

$$= (1 - e) - (0 - 1) + (2 \ln 2 - 2) - (0 - 1) + (e - 2)$$

$$- (2 - 2 \ln 2) + (3 - e) - (0 - 1) + \dots$$

$$= 65 + 4 \ln 2 - 7e \quad \therefore a - b^3 - c = 8$$

2. **Ans (2)**

$$(y^3 - 2x^2y) dx + (2xy^2 - x^3) dy = 0$$

Multiply by $2xy$

$$y^2(y^3 - 2x^2y)2xdx + x^2(2y^2 - x^3)2ydy = 0$$

Let $x^2 = u$, $y^2 = v$

$$\therefore v(v - 2u)du + u(2v - u)dv = 0$$

$$(v^2du + 2vudv) - (u^2dv + 2uvdu) = 0$$

$$d(uv^2) = d(u^2v)$$

$$uv^2 = u^2v + c$$

$$\therefore \boxed{x^2y^4 = x^4y^2 + c}$$

Put $x = 1$, $y = 2$

$$16 = u + c \therefore c = 12$$

$$x^2y^4 = x^4y^2 + 12$$

Now put $x = 2$

$$4y^4 = 16y^2 + 12$$

$$y^4 - 4y^2 - 3 = 0, \quad y^2 = \frac{4 \pm \sqrt{16 + 12}}{2}$$

$$y^2 = 2 \pm \sqrt{7} \therefore [|y(2)|] = 2$$

3. **Ans (3)**

$$x \cdot (f'(x) + f''(x)) = -f'(x) - e^{-x}$$

$$\int \frac{x}{1} \cdot \frac{e^x(f'(x) + f''(x))}{1} = -e^x(f'(x) - 1)$$

Integrate both side

$$xe^xf'(x) - \int e^xf'(x) dx = - \int e^xf'(x) dx - x + c$$

$$xe^xf'(x) = c - x \quad \text{put } x = 0 \therefore c = 0$$

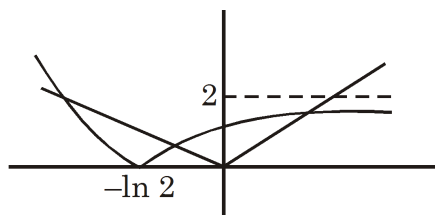
$$f'(x) = -e^{-x} \therefore f(x) = e^{-x} + \lambda \therefore \lambda = -2$$

$$f(x) = e^{-x} - 2$$

$$|f(x)| = |x|$$

$$|e^{-x} - 2| = |x|$$

$\therefore$ No. of solution is 3.



4. **Ans (0)**

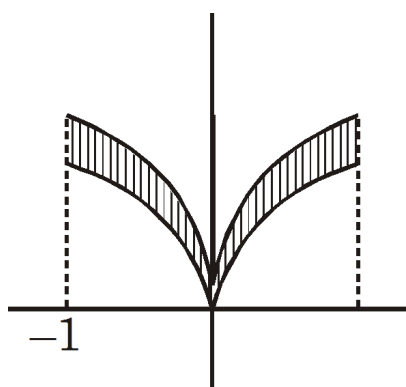
$$f(x) = \sin^{-1} \left(\frac{2|x|}{1+x^2} \right) + \sin^{-1} \left(\frac{|x|}{\sqrt{1+x^2}} \right)$$

$$\text{Put } |x| = \tan \theta, \quad \theta \in \left[0, \frac{\pi}{2} \right)$$

$$\therefore f(x) = \sin^{-1}(\sin 2\theta) + \sin^{-1}(\sin \theta)$$

$$= \begin{cases} 2\theta + \theta, & \theta \in \left[0, \frac{\pi}{4} \right] \\ \pi - 2\theta + \theta & \theta \in \left(\frac{\pi}{4}, \frac{\pi}{2} \right) \end{cases}$$

$$\therefore f(x) = 3\tan^{-1}|x|, \quad x \in [-1, 1]$$



$$\text{Area} = 2 \int_0^1 (3\tan^{-1}x - \tan^{-1}x) dx$$

$$= 4 \int_0^1 \tan^{-1}x dx$$

$$= \pi - \ln 4$$

$$\therefore [b - a] = [4 - \pi] = 0$$

5. Ans (1)

$$I_n = \int_1^{\pi/4} (\tan x)^n dx$$

$$\text{then } I_n + I_{n-2} = \frac{1}{n-1}$$

$$\Rightarrow I_n + I_{n+2} = \frac{1}{n+1}$$

$$\text{and } I_{n+1} + I_{n+3} = \frac{1}{n+2}$$

$$\Rightarrow S_n = (I_n + I_{n+2}) (I_{n+1} + I_{n+3}) = \frac{1}{(n+1)(n+2)}$$

$$\text{Now } \lim_{n \rightarrow \infty} \sum S_n = 1$$

6. Ans (6)

$$\int \frac{2x^4 + x^3 + x}{(x^3 + x^2 + x - 1)^3} dx = \int \frac{(2x^4 + x^3 + x)dx}{x^3 \left(x^2 + x + 1 - \frac{1}{x}\right)^3}$$

$$= \int \frac{\left(2x + 1 + \frac{1}{x^2}\right) dx}{\left(x^2 + x + 1 - \frac{1}{x}\right)^3} \quad \text{let } x^2 + x + 1 - \frac{1}{x} = t$$

$$= \int \frac{dt}{t^3} = \frac{-1}{2\left(x^2 + x + 1 - \frac{1}{x}\right)^2} + c$$

$$\therefore A = -2 \quad \text{and} \quad f(x) = x^2 + x + 1 - \frac{1}{x}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

PART TEST

04-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	D	A	D	C	C	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	B	C	A	B	D	A	D	C	A
	Q.	1	2	3	4	5	6	7	8	9	10
A.	0.57 to 0.58	7.44 to 7.46	26.62 to 26.64	10.00	3.50	4.73 to 4.76	32.00	260.00	0.60	5.00	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	3	D	A	A	A	B	C	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	C	B	B	D	D	D	A	C	D
	Q.	1	2	3	4	5	6	7	8	9	10
A.	4.00	11.00	2.00	25.00	100.00	80.00	3.00	5.00	5.00	1.00	

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	B	A	A	A	A	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	C	A	B	C	A	A	C	A	A
	Q.	1	2	3	4	5	6	7	8	9	10
A.	0.00	8.00	10.20	2.00	4.50	1.00	20.00	9.00	14.00	0.50	

HINT – SHEET
PART-1 : PHYSICS
SECTION-I
1. Ans (C)

Although all the four functions are written in the form $f(ax \pm bt)$, only (c) among the four functions is finite everywhere at all times. Hence only (c) represents a wave.

2. Ans (D)

Reflection from the rigid end gives a phase shift of 180° .

3. Ans (C)

$$P = \frac{1}{2} \mu A^2 \omega^2 v$$

4. Ans (D)

$$\xi = A \sin(kx - \omega t)$$

$$P_{\text{ex}} = -B \frac{d\xi}{dx} = -BAk \cos(kx - \omega t)$$

$$\text{amplitude of } P_{\text{ex}} = BAk$$

$$= (5 \times 10^5) (10^{-4}) = BAk$$

$$= (5 \times 10^5) (10^{-4}) \left(\frac{2\pi}{0.2} \right) = 5\pi \times 10^2 \text{ Pa}$$

So correct ans is (D)

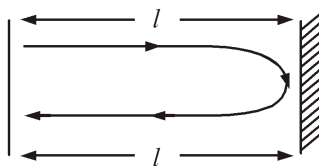
5. Ans (A)

"Loudness is determined by the amplitude of sound and pitch by the frequency of sound."

Is correct statement.

6. **Ans (D)**

Drumming frequency = 40 cycle/min = 40 cycle/60 sec



$$\text{Drumming time period} = \frac{1}{f} = \frac{60 \text{ sec}}{40 \text{ cycle}}$$

$$= \frac{3}{4} \text{ sec/cycle}$$

(time duration between consecutive drumming)

During this time interval, if sound goes to mountain and comes back then echo will not be heard distinctly.

$$\frac{3}{4} = \frac{2\ell}{v} \quad \dots(1)$$

Now, if he moves 90 m. This situation arises at $t = 60 \text{ cycle/min}$, $T = 1/f = 1 \text{ sec/cycle}$

$$\text{so for this case } 1 = \frac{2(\ell + 90)}{v} \quad \dots(2)$$

Solving equation 1 and 2

$$\text{set } \ell = 270 \text{ m}$$

7. **Ans (C)**

At path difference $\frac{\lambda}{6}$, phase difference is $\frac{\pi}{3}$

$$I = I_0 + I_0 + 2I_0 \cos \frac{\pi}{3} = 3 I_0$$

$$I_{\max} = 4I_0$$

$$\text{So the required ratio is } \frac{3 I_0}{4 I_0} = 0.75.$$

8. **Ans (C)**

$$d = 100 \lambda$$

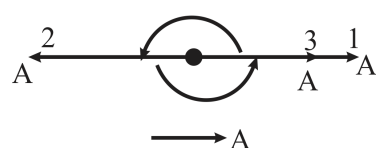
100 bright fringes on either sides of central bright fringes

$$\therefore \text{total} = 199$$

9. **Ans (C)**

The point at 1 m below O corresponds to central maxima.

10. **Ans (C)**



$$\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x$$

$$= \pi$$

11. **Ans (A)**

$$T_1 = 2\pi \sqrt{\frac{m}{k}}$$

$$T_2 = 2\pi \sqrt{\frac{m}{\frac{k}{2}}}$$

$$T_3 = 2\pi \sqrt{\frac{m}{2k}}$$

12. **Ans (B)**

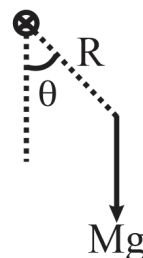
$$\text{T.E. of SHM} = \frac{1}{2} kA^2$$

$$\text{P.E.} = \frac{1}{2} kx^2 = \frac{1}{2} k \left(\frac{A}{2} \right)^2$$

$$2.5 = \frac{1}{2} k \frac{A^2}{4}$$

$$\Rightarrow \frac{1}{2} kA^2 = 10 \text{ J}$$

13. **Ans (C)**



$$\tau = -MgR \sin \theta$$

$\therefore \theta$ is small

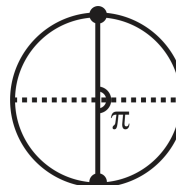
$$\tau = -MgR\theta$$

$$\Rightarrow c = MgR$$

$$T = 2\pi \sqrt{\frac{3MR^2}{2C}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{3R}{2g}}$$

14. **Ans (A)**



At maximum separation their line joining must be vertical.

So from phasor diagram.

15. **Ans (B)**

$$A = A_0 e^{-\left(\frac{b}{2m}\right)t}$$

$$\frac{A_0}{2} = A_0 e^{-\frac{b}{2m}(1 \text{ min})}$$

$$\frac{1}{2} = e^{-\frac{b}{2m}(1 \text{ min})}$$

$$-\frac{b}{2m}(1 \text{ min}) = -\ln 2 \dots (1)$$

$$\text{Now } A = A_0 e^{-\frac{b}{2m}(3 \text{ min})}$$

$$A = A_0 e^{-3 \ln 2}$$

$$\therefore A = 2^{-3} A_0$$

16. **Ans (D)**

"If the driving frequency is equal to the frequency of the system then amplitude of the system will be maximum"

17. **Ans (A)**

Correct Answer is

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

18. **Ans (D)**

Correct Answer is

"Sound waves are of longer wavelength"

19. **Ans (C)**

$$a = 0.3 \text{ mm}$$

$$D = 40 \times 10^{-52} \text{ m}$$

$$\frac{\lambda D}{a} = 0.8 \times 10^{-3}$$

$$\lambda = \frac{0.8 \times 10^{-3} \times 0.3 \times 10^{-3}}{40 \times 10^{-2}}$$

$$= \frac{24}{4} \times 10^{-7}$$

$$= 6 \times 10^{-7}$$

$$= 6000 \times 10^{-10} \text{ m}$$

$$= 6000 \text{ Å}$$

20. **Ans (A)**

Brewster's angle = 53°

$$\tan 53^\circ = \frac{n_d}{n_r} = \frac{4}{3}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (0.57 to 0.58)**

$$I = I_0 \cos^2 \theta$$

$$\Rightarrow \frac{I_0}{3} = I_0 \cos^2 \theta$$

$$\therefore \theta = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

2. **Ans (7.44 to 7.46)**

$$\text{R.P.} = \frac{a}{1.22\lambda}$$

$$= \frac{50 \times 10^{-2}}{1.22 \times 550 \times 10^{-9}}$$

$$= 7.45$$

3. **Ans (26.62 to 26.64)**

$$\frac{I}{C} = \frac{1}{2} \epsilon_0 E^2$$

$$\frac{5 \times 10^{-2}}{3 \times 10^8} = \frac{1}{2} \times 8.85 \times 10^{-12} \times E^2$$

$$E = \frac{5 \times 2}{3 \times 8.85} \times 10^2$$

$$= \frac{1000}{3 \times 8.05} = 37.66$$

$$E_{\text{rms}} = 26.63$$

4. **Ans (10.00)**

$$\frac{\lambda}{2} = 5$$

$$\Rightarrow \lambda = 10 \text{ m}$$

5. **Ans (3.50)**

$$L_2 P - L_1 P = \frac{\lambda}{2}$$

6. **Ans (4.73 to 4.76)**

$$\frac{I}{C} = \frac{1}{2} \epsilon_0 E^2$$

$$E = \sqrt{\frac{2I}{\epsilon_0 C}} = \sqrt{\frac{2 \times 3 \times 10^{14}}{8.85 \times 10^{-12} \times 3 \times 10^8}}$$

$$= \sqrt{\frac{2}{8.85}} \times 10^9$$

$$= 0.474 \times 10^9$$

$$= 4.74 \times 10^8 \text{ N/C}$$

7. **Ans (32.00)**

Intensity after crossing first polariser

$$= \frac{I}{2}$$

Intensity after crossing second polariser

$$= \frac{I}{2} \cos^2 \frac{\pi}{3} = \frac{I}{8}$$

Intensity after crossing third polariser

$$= \frac{I}{8} \cos^2 \frac{\pi}{3} = \frac{I}{32}$$

8. **Ans (260.00)**

For first minima,

$$a \sin \theta_1 = \lambda_1$$

$$\sin \theta_1 = \frac{\lambda_1}{a} = \frac{650 \text{ nm}}{a}$$

for second maxima,

$$a \sin \theta_2 = \frac{5\lambda}{2}$$

for coinciding,

$$\sin \theta_1 = \sin \theta_2$$

$$\lambda_2 = 260 \text{ nm}$$

9. **Ans (0.60)**

$$v = v_0 \sqrt{\frac{c-v}{c+v}}$$

$$\frac{1}{4} = \frac{c-v}{c+v}$$

$$4c - 4v = c + v$$

$$3c = 5v$$

$$v = \frac{3}{5} c = 0.6$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (B)**

$$\frac{\text{Radial probability density at } r = a_0}{\text{Radial probability density at } r = 0} = \frac{R^2(a_0)}{R^2(0)}$$

$$\text{For 1s orbital : } R(r) = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{-r/a_0}$$

$$\Rightarrow \frac{R^2(a_0)}{R^2(0)} = \frac{(1/\pi a_0^3) e^{-2r/a_0}}{(1/\pi a_0^3) e^0} = e^{-2}$$

2. **Ans (3)**

For zero order reaction, $kt = [A_0] - [A]$

$$(i) [A] = [A_0] - kt$$

$$= 10 - (2 \times 10^{-6}) \times (2 \times 24 \times 60) = 9.994 \text{ M}$$

$$(ii) \text{ when } t = t_{1/2}, [A] = \frac{[A_0]}{2}$$

Hence,

$$t_{1/2} = \frac{[A] - \frac{[A_0]}{2}}{k} = \frac{[A_0]}{2k} = \frac{10}{2 \times (2 \times 10^{-6})}$$

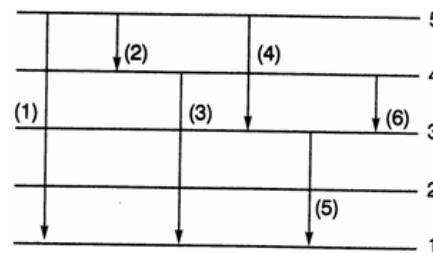
$$= 2.5 \times 10^6 \text{ min}$$

(iii) When reaction is complete, $[A] = 0$

$$\therefore t = \frac{[A_0]}{k} = \frac{10}{2 \times 10^{-6}}$$

$$= 5 \times 10^6 \text{ min}$$

3. **Ans (D)**



Total radiations are = 6

4. **Ans (A)**

Corrected C^{14} content :

Let initial C^{14} count = A_0 ;

$$A_0 + A_0/10 = 15.4; A_0 = 14$$

$$t = \frac{1}{\lambda} \log \frac{14}{7} = 5.770 \times 10^3 \text{ years } t = \frac{\ln 2}{\lambda}$$

$$= t_{1/2}; t = 5770 \text{ years}$$

5. **Ans (A)**

If the motion of nucleus is considered electron and nucleus will revolve round the center of mass of the atom such that the distance between them remains always constant. Now,

From the concept of center of mass,

$$m_e \cdot r_1 = m_n \cdot r_2 = m_n \cdot (r - r_1)$$

$$\therefore r_1 = \frac{m_n}{m_n + m_e} \cdot r$$

$$\text{and } r_2 = \frac{m_e}{m_n + m_e} \cdot r$$

Now, from Bohr's model, if nucleus is also revolving, the total angular momentum,

$$(I\omega)_e + (I\omega)_n = \frac{nh}{2\pi}$$

$$\therefore m_e r_1^2 \omega + m_n r_2^2 \omega = \frac{nh}{2\pi}$$

$$\therefore m_e \left(\frac{m_n}{m_n + m_e} \cdot r \right)^2 \omega + m_n \left(\frac{m_e}{m_n + m_e} \cdot r \right)^2 \omega = \frac{nh}{2\pi}$$

$$\therefore \left(\frac{m_e \cdot m_n}{m_n + m_e} \right) r^2 \omega = \frac{nh}{2\pi}$$

$$\therefore \mu r^2 \omega = \frac{nh}{2\pi} \dots (1)$$

$$\text{Where, } \mu = \frac{m_e \cdot m_n}{m_n + m_e} = \text{reduced mass}$$

Now, when nucleus is considered is stationary,

$$m_e r^2 \omega = \frac{nh}{2\pi} \dots (2)$$

From eq. (1) and (2), if the motion of nucleus is considered, in all the calculations regarding electron, the mass of electron should be replaced by the reduced mass.

6. **Ans (A)**

$$\text{at time } \lambda = \frac{0.693}{6.93 \times 10^9} \Rightarrow 10^{-10} \text{ years}^{-1}$$

$$t = \frac{2.303}{10^{-10}} \log \left(\frac{w_0}{w_0 - x} \right)$$

$$t = \frac{2.303}{10^{-10}} \log(8)$$

$$= 2.079 \times 10^{10} \text{ years}$$

7. **Ans (B)**

From question, $\Delta x = m \cdot \Delta v$

Now, from Heisenberg's uncertainty principle :

$$\Delta x \cdot m \cdot \Delta v \geq \frac{h}{4\pi}$$

$$\therefore (m \cdot \Delta v)^2 \geq \frac{h}{4\pi}$$

$$\therefore (m \cdot \Delta v)_{\min} = \sqrt{\frac{h}{4\pi}}$$

$$\therefore v_{\min} = \frac{1}{2m} \sqrt{\frac{h}{\pi}}$$

$$= \frac{1}{2 \times (9.109 \times 10^{-31})} \sqrt{\frac{6.626 \times 10^{-34}}{3.14}}$$

$$= 7.97 \times 10^{12} \text{ m/s}$$

8. **Ans (C)**

Follow the rules for stability of resonating structures

(structure with more number of π -bonds is more stable)

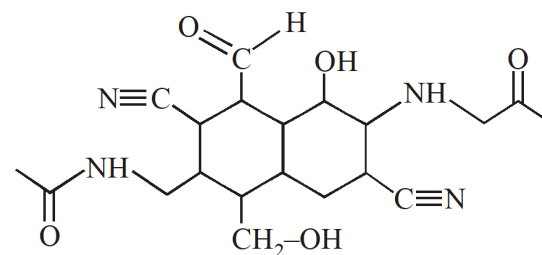
9. **Ans (A)**

oxygen is more electronegative than sulphur hence alcohol will form stronger than thiol.

10. **Ans (D)**

$\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH}_2$ has two α -hydrogen for hyperconjugation.

11. **Ans (D)**



Total functional groups : -

- (a) aldehyde (b) ketone
- (c) 2°-amine (d) cyanides
- (e) alcohol (f) substituted amides
- (g) isocyanide

12. **Ans (C)**

pK_a represent more tendency to donate lone pair
If lone pair involving resonance within the molecular less tends to donate lone pair to outside.

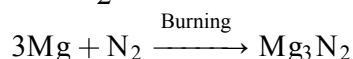
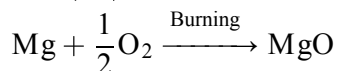
13. **Ans (B)**

$C_1 - C_2$ bond is shorter than $C_3 - C_4$ bond
Because $C_1 - C_2$ bond less participation with double bond,
then $C_3 - C_4$

14. **Ans (B)**

Due to ortho effect and -I and -M effect of $-\text{NO}_2$ group.

15. **Ans (D)**

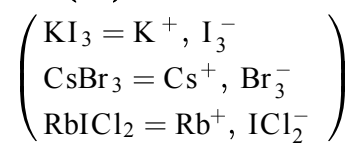


16. **Ans (D)**

BeH_2 exists in polymeric form ($3c-2e^-$ bonding)



17. **Ans (D)**



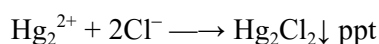
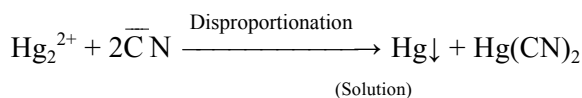
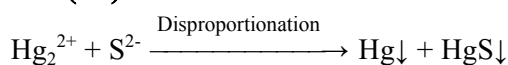
POLYHALIDE IONS



18. **Ans (A)**

Unpaired electrons = $\text{V}^{2+} < \text{Cr}^{2+} = \text{Fe}^{2+} < \text{Mn}^{2+}$
 $\text{Sc} < \text{Ti} < \text{Cr} < \text{Mn}$: number of oxidation states
 $\text{Ni}^{2+} < \text{Cr}^{2+} < \text{Fe}^{2+} < \text{Mn}^{2+}$: ionic size

19. **Ans (C)**



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (4.00)**

More electro negative element like F O N shows generally -I effect.

Here carbon in CH_3 is electro positive,
There fore is shows +I effect.

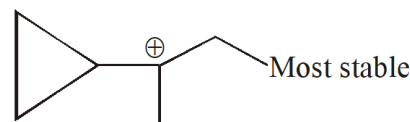
2. **Ans (11.00)**

DBE = Number of π -bond (unsaturation) + Number of cyclic ring present = $9 + 2 = 11$

π bond of cyanide group $\text{R}-\text{C}\equiv\text{N}$, aldehydic group

$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ & ketonic group $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R}$ are also considered.

3. **Ans (2.00)**



4. **Ans (25.00)**

Moseley's law is

$$\sqrt{\nu} = \sqrt{\frac{c}{\lambda}} = a(Z - b)$$

For copper,

$$\sqrt{\frac{c}{15.42 \times 10^{-9}}} = a(29 - b) \quad \dots(1)$$

For molybdenum

$$\sqrt{\frac{c}{7.12 \times 10^{-9}}} = a(42 - b) \quad \dots(2)$$

and for impurity,

$$\sqrt{\frac{c}{22.85 \times 10^{-9}}} = a(Z - b) \quad \dots(3)$$

From eq. (1) and (2),

$$a = 5.049 \times 10^{-6} \text{ s}^{-\frac{1}{2}}, \quad b = 1.375,$$

Putting these values eq. (3),

the atomic number of impurity, $Z = 24.07 = 24$ (approx)

5. **Ans (100.00)**

From question, rate of uncatalysed reaction at 500 K

= rate catalysed reaction at 400 K

$$K_{500} = K'_{400}$$

$$\therefore A \cdot e^{-\frac{E_a}{R \times 500}} = A \cdot e^{-\frac{E_a}{R \times 400}}$$

$$\therefore \frac{E_a}{500} = \frac{E_a'}{400}$$

$$\therefore \frac{E_a}{500} = \frac{E_a - 20}{400}$$

$$\therefore E_a = 100 \text{ kJ/mol}$$

6. **Ans (80.00)**

$$e^{-\frac{E_a}{RT}} \times 100 = 3.8 \times 10^{-16}; e^{-\frac{E_a}{RT}} = 3.8 \times 10^{-18}$$

$$-\frac{E_a}{RT} = \ln(3.8 \times 10^{-18});$$

$$R = 2 \text{ and } T = 1000$$

$$E_a = 80 \text{ kJ/mol}$$

7. **Ans (3.00)**

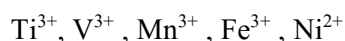
K, Rb and Cs produce superoxide ($M^+O_2^-$) on reaction with O_2 .

8. **Ans (5.00)**

5, All nitrates of alkali earth metal.

9. **Ans (5.00)**

Total five ions i.e.



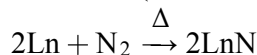
10. **Ans (1.00)**

[1 $\rightarrow$ F]

Th Z = 92 (1st element of 5f series)

Ti Z = 22 (1st element of 3d series)

Lu Z = 71 (Last element of 4f series)



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

$$\int \frac{\cos^2 x - 2005}{\cos^{2005} x} dx$$

$$= \int (\cos x)^{-2005} \times \sec^2 x dx - 2005 \int \frac{dx}{\cos^{2005} x}$$

$$= (\cos x)^{-2005} \times (-\cot x) - \int (-2005)(\cos x)^{-2006} \times (-\sin x)$$

$$(-\cot x) dx - 2005 \int \frac{dx}{\cos^{2005} x}$$

$$= -\frac{\cot x}{(\cos x)^{2005}} + C$$

2. **Ans (B)**

Equation of tangent is $y - \beta = f'(\alpha) \{x - \alpha\}$

$$\text{Also, } \alpha = \frac{\beta - \alpha f'(\alpha)}{\sqrt{f'(\alpha)^2 + 1}}$$

$$\text{Thus, } \frac{dy}{dx} = \frac{1}{2} \left(\frac{y^2 - x^2}{xy} \right)$$

Thus, equation of curve is $x^2 + y^2 - 2x = 0$

Then the required length of intercept is 2.

3. **Ans (B)**

Number of elements in $A \times B = 2 \times 4 = 8$

$\therefore$ Required number of subsets

$$= {}^8C_3 + {}^8C_4 + \dots + {}^8C_8$$

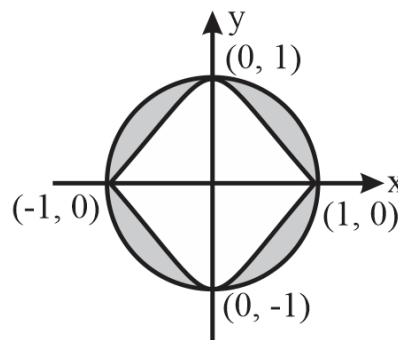
$$= ({}^8C_0 + {}^8C_1 + \dots + {}^8C_8) - ({}^8C_0 + {}^8C_1 + {}^8C_2)$$

$$= 2^8 - (1 + 8 + 28) = 219$$

4. **Ans (A)**

$$\text{Required area} = \pi - 4 \left[\int_0^1 (1 - x^2) dx \right]$$

$$= \pi - \frac{8}{3}$$



5. **Ans (A)**

$$I_n = \int_0^\pi e^x (\sin x)^n dx,$$

$$= e^x (\sin x)^n \Big|_0^\pi - n \int_0^\pi (\sin x)^{n-1} \cos x e^x dx$$

$$I_n = 0 - n \left[(\sin x)^{n-1} \cos x \cdot e^x \Big|_0^\pi \right]$$

$$- \int_0^\pi \left[(n-1)(\sin x)^{n-2} \cos^2 x - (\sin x)^n \right] e^x dx$$

$$\Rightarrow I_n = n \left[(n-1) \int_0^\pi \sin^{n-2} (1 - \sin^2 x) e^x dx - \int_0^\pi (\sin x)^n e^x dx \right]$$

$$\Rightarrow I_n = n[(n-1)I_{n-2} - (n-1)I_n - I_n]$$

$$\Rightarrow I_n = (n^2 - n)I_{n-2} - n^2 I_n$$

$$\Rightarrow I_n = \frac{n(n-1)I_{n-2}}{1 + n^2}$$

$$\therefore \frac{I_3}{I_1} = \frac{3 \times 2}{10} = \frac{3}{5}$$

6. **Ans (A)**

$$\text{Let } P(n) = \frac{n^5}{5} + \frac{n^3}{3} + \frac{7n}{15}$$

$$P(1) = \frac{1}{5} + \frac{1}{3} + \frac{7}{15} = \frac{3+5+7}{15} = 1$$

∴ (c), (d) are ruled out.

$$P(2) = \frac{32}{5} + \frac{8}{3} + \frac{14}{15}$$

$$= \frac{96+40+14}{15} = \frac{150}{15}$$

$$= 10 = \text{An even integer.}$$

(can use binomial theorem also)

7. **Ans (A)**

$$x dy - y dx = \sqrt{x^2 - y^2} dx$$

$$\Rightarrow d\left(\frac{y}{x}\right) = \frac{\sqrt{1 - \left(\frac{y}{x}\right)^2}}{x} dx$$

$$\Rightarrow \int \frac{d\left(\frac{y}{x}\right)}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} = \int \frac{dx}{x}$$

$$\Rightarrow \sin^{-1} \frac{y}{x} = \ln x + \ln c \text{ or } Cx = Cx = e^{\sin^{-1} \frac{y}{x}}$$

8. **Ans (B)**

$$I = \int \frac{(x^2 + 1)}{(x^4 - x^2 + 1) \cot^{-1} \left(x - \frac{1}{x}\right)} dx$$

$$= \int \frac{\left(1 + \frac{1}{x^2}\right)}{\left(x^2 + \frac{1}{x^2} - 2 + 1\right) \cot^{-1} \left(x - \frac{1}{x}\right)} dx$$

$$\text{put } x - \frac{1}{x} = t$$

$$\Rightarrow \left(1 + \frac{1}{x^2}\right) dx = dt$$

$$I = \int \frac{dt}{(t^2 + 1) \cot^{-1} t}$$

$$\text{Let } \cot^{-1} t = u$$

$$\Rightarrow -\frac{1}{(1+t^2)} dt = du$$

$$I = \int \frac{-du}{u} = -\ln|u| + C$$

$$= -\ln \left| \cot^{-1} \left(x - \frac{1}{x}\right) \right| + C$$

9. **Ans (B)**

Now we have to find x, y such that both are integers and $x^2 + y^2 = 64$

$$\text{If } x = 0, y = \pm 8$$

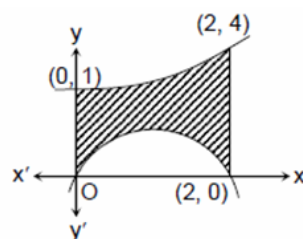
$$y = 0, x = \pm 8$$

$$R = \{(0, 8), (0, -8), (8, 0), (-8, 0)\}$$

10. **Ans (A)**

Required area

$$= \int_0^2 (2^x - (2x - x^2)) dx$$



11. **Ans (D)**

$$\int_0^{1.5} x[x^2] dx = \int_0^1 x[x^2] dx + \int_1^{\sqrt{2}} x[x^2] dx + \int_{\sqrt{2}}^{1.5} x[x^2] dx$$

$$= \int_0^1 0 dx + \int_1^{\sqrt{2}} x \cdot 1 dx + \int_{\sqrt{2}}^{1.5} x \cdot 2 dx$$

$$= \left| \frac{x^2}{2} \right|_1^{\sqrt{2}} + 2 \left| \frac{x^2}{2} \right|_{\sqrt{2}}^{1.5}$$

$$= \frac{2}{2} - \frac{1}{2} + \left(\frac{9}{4} - 2 \right) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$[\because 0 < x < 1 \Rightarrow 0 < x^2 < 1 \Rightarrow [x^2] = 0]$$

$$1 < x < \sqrt{2} \Rightarrow 1 < x^2 < 2 \Rightarrow [x^2] = 1$$

$$\sqrt{2} < x < 1.5 \Rightarrow 2 < x^2 < 2.25 \Rightarrow [x^2] = 2.$$

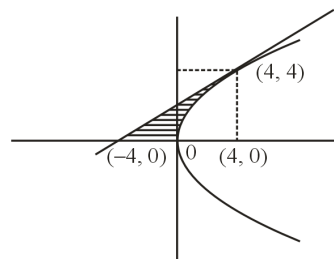
12. **Ans (C)**

Tangent is $4y = 2(x + 4)$ or $2y = x + 4$

$$\text{Area} = \frac{1}{2} \cdot 8 \cdot 4 - \frac{2}{3} \cdot 16$$

$$= 16 - \frac{32}{3}$$

$$= \frac{16}{3}$$



13. **Ans (A)**

$$\begin{aligned} x^2(y+1)dx + y^2(x-1)dy &= 0 \\ \text{or } \frac{x^2 dx}{x-1} &= -\frac{y^2 dy}{y+1} \\ \text{or } \int \left[x+1 + \frac{1}{x-1} \right] dx &= -\int \left[y-1 + \frac{1}{y+1} \right] dy \\ \text{or } \frac{x^2}{2} + x + \ln(x-1) &= -\left[\frac{y^2}{2} - y + \ln(y+1) \right] + \ln C \\ \Rightarrow x^2 + y^2 + 2(x-y) + 2 \ln \frac{(x-1)(y+1)}{C} &= 0 \end{aligned}$$

14. **Ans (B)**

p	q	$\sim p$	$\sim p \vee q$	$p \Rightarrow p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Since $p \Rightarrow \sim p \vee q$ is false.

$\therefore$ truth values of p and q are T, F

15. **Ans (C)**

$$\begin{aligned} \int \frac{x^2}{(x^2+1)(x^2+4)} dx \\ = \frac{1}{3} \int \left[\frac{4}{x^2+4} - \frac{1}{x^2+1} \right] dx \\ = -\frac{1}{3} \int \frac{1}{x^2+1} dx + \frac{4}{3} \int \frac{1}{x^2+4} dx \\ = -\frac{1}{3} \tan^{-1} x + \frac{4}{3} \times \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C \\ = -\frac{1}{3} \tan^{-1} x + \frac{2}{3} \tan^{-1} \left(\frac{x}{2} \right) + C \end{aligned}$$

16. **Ans (A)**

$$\begin{aligned} I_1 &= \int_{-100}^{101} \frac{dx}{(5+2x-2x^2)(1+e^{2-4x})} \\ &= \int_{-100}^{101} \frac{dx}{(5+2(1-x)-2(1-x^2))(1+e^{2-4(1-x)})} \\ \Rightarrow 2I_1 &= \int_{-100}^{101} \frac{dx}{5+2x-2x^2} = I_2 \\ \Rightarrow \frac{I_1}{I_2} &= \frac{1}{2} \end{aligned}$$

17. **Ans (A)**

Since A.M. of n observations is M.

$\therefore$ Total of n observations = Mn

sum of (n-4) observations = a

$\therefore$ Sum of remaining observations = Mn - a

$\therefore$ Required Mean = $\frac{Mn - a}{4}$

18. **Ans (C)**

$$\begin{aligned} I_n &= x(\ln x)^n - \int \frac{(n)(\ln x)^{n-1}}{x} \cdot x dx \\ &= x(\ln x)^n - n I_{(n-1)} \\ \text{or } I_n + n I_{n-1} &= x(\ln x)^n \end{aligned}$$

19. **Ans (A)**

$$\begin{aligned} &\sim[(\sim p) \wedge q] \\ &= \sim(\sim p) \vee (\sim q) \\ &= p \vee (\sim q) \end{aligned}$$

20. **Ans (A)**

$$\begin{aligned} \int \sin x \log \left(\cot \frac{x}{2} \right) dx \\ I &= -\cos x \log \left(\cot \frac{x}{2} \right) - \int \frac{1}{\cot \frac{x}{2}} \cdot \left(\frac{-\cos x}{2} \right) dx \\ &= (-\cos x) dx \\ &= -\cos x \log \left(\cot \frac{x}{2} \right) - \frac{1}{2} \int \frac{\cos x}{\cos x/2 \sin x/2} dx \\ &= -\cos x \log \left(\cot \frac{x}{2} \right) - \frac{1}{2} \int \frac{(\cos^2 x/2 - \sin^2 x/2)}{\cos x/2 \sin x/2} dx \\ &= -\cos x \log \left(\cot \frac{x}{2} \right) - \frac{1}{2} \int \left[\cot \frac{x}{2} dx - \tan \frac{x}{2} dx \right] \\ &= -\cos x \log \left(\cot \frac{x}{2} \right) - \frac{1}{2} \left[2 \log \left(\sin \frac{x}{2} - 2 \log \sec \frac{x}{2} \right) \right] + C \\ &= -\cos x \log \left(\cot \frac{x}{2} \right) - \log \left(\sin \frac{x}{2} \right) + \log \left(\sec \frac{x}{2} \right) + C \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (0.00)**

Put

$$y = x + 2$$

2. Ans (8.00)

The given limit is

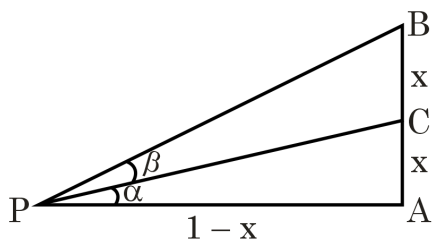
$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left[\frac{1}{na} + \frac{1}{na+1} + \frac{1}{na+2} + \dots + \frac{1}{na+n(b-a)} \right] \\ &= \lim_{n \rightarrow \infty} \sum_{r=0}^{(b-a)n} \frac{1}{na+r} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{(b-a)n} \frac{1}{a+r/n} \\ &= \int_0^{(b-a)} \frac{dx}{a+x} = [\log(a+x)]_0^{b-a} \\ &= \log b - \log a = \log(b/a) \end{aligned}$$

3. Ans (10.20)

Let $\angle APC = \alpha$, $AC = CB = x$.

It is given that $AP = 5AB$

$$\Rightarrow AP = 10x.$$



In $\triangle APC$, we have $\tan \alpha = \frac{AC}{AP}$

$$\Rightarrow \tan \alpha = \frac{x}{10x} = \frac{1}{10}$$

From $\triangle APB$, we have

$$\tan(\alpha + \beta) = \frac{AB}{AP}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{2x}{10x}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{1}{5}$$

Now, $\tan \beta = \tan[(\alpha + \beta) - \alpha]$

$$= \frac{\tan(\alpha + \beta) - \tan \alpha}{1 + \tan(\alpha + \beta) \tan \alpha}$$

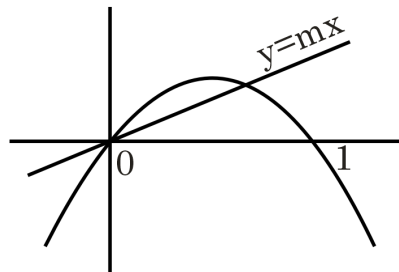
$$\Rightarrow \tan \beta = \frac{(1/5) - (1/10)}{1 + (1/5)(1/10)} = \frac{5}{51}$$

$$\cot \beta = \frac{51}{5}$$

4. Ans (2.00)

Solving $y = x - x^2$ and $y = mx$, we get the points of intersection $(0, 0)$ and $[1 - m, m(1 - m)]$

Case I : $m < 1$

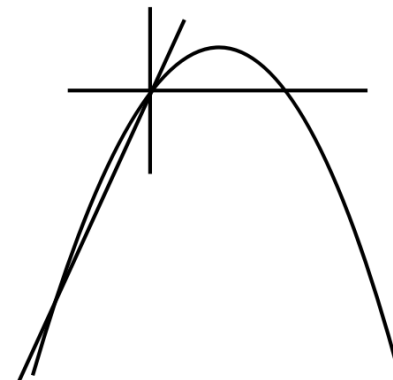


$$\text{Shaded area} = \frac{9}{2}$$

$$\Rightarrow \int_0^{1-m} (x - x^2) dx - \int_0^{1-m} mx dx = \frac{9}{2}$$

Solving this, we get $m = -2$

Case II : $m > 1$



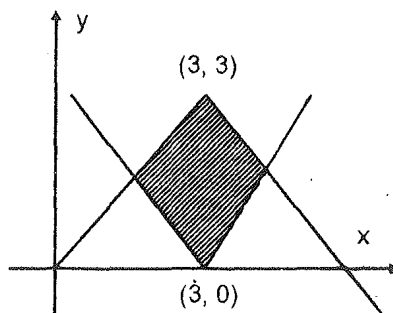
$$\text{Shaded area} = \frac{9}{2}$$

$$\int_{1-m}^1 (x - x^2 - mx) dx = \frac{9}{2}$$

solving this, we get $m = 4$

5. Ans (4.50)

$$\text{Required area} = \frac{3}{\sqrt{2}} \times \frac{3}{\sqrt{2}} = \frac{9}{2}$$



6. **Ans (1.00)**

$$\begin{aligned}\frac{dy}{dx} &= \frac{y^3 + y^2 \sqrt{y^2 - x^2}}{x^3} \\ \text{put } \frac{y}{x} &= v \\ \Rightarrow v + x \frac{dv}{dx} &= v^3 + v^2 \sqrt{v^2 - 1} \\ \Rightarrow \frac{dv}{v(v^2 - 1) + v^2 \sqrt{v^2 - 1}} &= \frac{dx}{x} \\ \Rightarrow \int \frac{dv}{v \sqrt{v^2 - 1} (\sqrt{v^2 - 1} + v)} &= \int \frac{dx}{x} \\ &= \log(v + \sqrt{v^2 - 1}) - \log v = \log x + C \\ \Rightarrow \log\left(\frac{y}{x} + \sqrt{\left(\frac{y}{x}\right)^2 - 1}\right) - \log \frac{y}{x} &= \log x + \log k \\ &= \log x + \log k \\ \therefore Cxy &= y + \sqrt{y^2 - x^2}\end{aligned}$$

7. **Ans (20.00)**

Here $n = 20$, Variance = 5

$$\Rightarrow \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2 = 5$$

If each observations is multiplied by 2,

Let $2x_i = y_i$

$$\begin{aligned}\therefore \text{New variance} &= \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2 \\ &= \frac{\sum (2x_i)^2}{n} - \left(\frac{\sum 2x_i}{n}\right)^2 \\ &= 4 \left[\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2 \right] = 4 \times 5 = 20\end{aligned}$$

8. **Ans (9.00)**

$$\begin{aligned}\frac{dy}{dx} + y^3(1 + \cos x) - \frac{y}{x} &= 0 \\ \frac{1}{y^3} \frac{dy}{dx} - \frac{1}{y^2 x} &= -(1 + \cos x) \\ \text{Let } \frac{-1}{y^2} &= u \\ \therefore \frac{1}{2} \frac{du}{dx} + \frac{u}{x} &= -(1 + \cos x) \\ \text{I. F.} &= e^{\int \frac{2}{x} dx} = x^2 \\ \therefore ux^2 &= -2 \int x^2 (1 + \cos x) dx \\ \Rightarrow \frac{x^2}{2y^2} &= \frac{x^3}{3} + x^2 \sin x + 2x \cos x - 2 \sin x + k\end{aligned}$$

9. **Ans (14.00)**

Here, powers of both $\cos x$ and $\sin x$ are odd positive integers ; therefore, put $z = \cos x$ or $z = \sin x$, but the power of $\cos x$ is greater.

Therefore, it is convenient to put $z = \cos x$

Let $z = \cos x$. Then $dz = -\sin x dx$

$$\begin{aligned}\int \sin^3 x \cos^5 x dx &= \int \sin^2 x \cos^5 x \sin x dx \\ &= \int (1 - \cos^2 x) \cos^5 x \sin x dx \\ &= \int (1 - z^2) z^5 (-dz) \\ &= - \left(\frac{z^6}{6} - \frac{z^8}{8} \right) + C \\ &= -\frac{\cos^6 x}{6} + \frac{\cos^8 x}{8} + C\end{aligned}$$

10. **Ans (0.50)**

$$\begin{aligned}\lim_{x \rightarrow \infty} \int_0^{\frac{x^2}{2}} \frac{t^2}{x^2(1+t^2)} dt \quad \left(\frac{\infty}{\infty} \text{ form} \right) \\ \Rightarrow \lim_{x \rightarrow \infty} \frac{\int_0^{\frac{x^2}{2}} \frac{t^2}{x^2(1+t^2)} dt}{x^2} \\ = \lim_{x \rightarrow \infty} \frac{\frac{x^4}{4+x^4} x}{2x} \\ = \lim_{x \rightarrow \infty} \frac{x^5}{2x^5 + 8} = \frac{1}{2}\end{aligned}$$